

IMAGINARY ELECTRIC COMPANY

Power Generation Division - Plant Integration

Alpilla 600 MW Combined Cycle Power Plant - 1 x IE-9H.02 multi-shaft power island

Heat Recovery Steam Generator IE-HR3 - Datasheet

|                 |                                                                                |
|-----------------|--------------------------------------------------------------------------------|
| Document number | IEC-ALP-DS-104                                                                 |
| Revision        | C                                                                              |
| Date            | 2026-12-04                                                                     |
| Status          | Revised for EPC engineering - supersedes Rev 0                                 |
| Addressee       | Istanbul EPC Muhendislik ve Taahhut A.S. (consortium leader) - EPC engineering |
| Contract        | ALP-EPC-001 (Owner contract); consortium agreement IEPC-IEC                    |

|                  |                                |                                |                                 |
|------------------|--------------------------------|--------------------------------|---------------------------------|
|                  | Prepared                       | Checked                        | Approved                        |
| Function         | IEC plant integration engineer | IEC lead engineer (discipline) | IEC project engineering manager |
| Signature / date | signed 2026-12-04              | signed 2026-12-04              | signed 2026-12-04               |

|     |            |                                             |
|-----|------------|---------------------------------------------|
| Rev | Date       | Description                                 |
| 0   | 2026-09-27 | Issued for EPC engineering (reference data) |
| 0   | 2026-09-27 | Issued for EPC engineering                  |
| A   | 2026-09-28 | TQ-IEC-016                                  |
| B   | 2026-09-28 | TQ-IEC-018: SCR for LDO                     |
| C   | 2026-12-04 | TQ-IEC-025: design temperatures             |

CASE STUDY - Imaginary Electric, its products and all data are fictional.

## 1. General

| Item                          | Data                                                                                                                                                                                                                         |
|-------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Type                          | IE-HR3 horizontal gas flow, triple-pressure reheat, natural circulation (HP drum thin-wall design with external cyclone separators), modular (22 modules), self-supporting casing, internally insulated                      |
| Design code                   | ASME Section I, PED 2014/68/EU module G (notified body)                                                                                                                                                                      |
| Cycling                       | 250 starts/year, 30-year fatigue life per EN 12952-3 (hot 180, warm 60, cold 10 per year)                                                                                                                                    |
| Gas side pressure loss at SRC | 30 mbar (incl. SCR and CO catalyst, clean)                                                                                                                                                                                   |
| Outlet to stack               | HRSG outlet transition duct flange, 9.2 m diameter (stack by EPC, IF-26)                                                                                                                                                     |
| SCR                           | V2O5/TiO2 honeycomb at 360 - 370 degC (between IP superheater and HP economiser 2), aqueous ammonia 24.5 % injection grid (IEC); design NOx at HRSG outlet 22 mg/Nm3 (15 % O2); NH3 slip <= 3 mg/Nm3; catalyst life 24,000 h |
| CO catalyst                   | Pt/Pd oxidation catalyst at 450 degC (after HP superheater/reheater)                                                                                                                                                         |
| Aqueous ammonia consumption   | 125 kg/h at base load on gas; 480 kg/h on LDO (GT-outlet NOx 150 mg/Nm3, 67 % removal, TQ-IEC-018)                                                                                                                           |
| Stack damper                  | not included (stack by EPC)                                                                                                                                                                                                  |

## 2. Thermal data at SRC-NG-100

**Rev B (TQ-IEC-018):** SCR catalyst volume increased by 35 % and ammonia injection grid resized for LDO operation with GT-outlet NOx up to 150 mg/Nm3 (water/fuel ratio 0.60); ammonium bisulphate protection: minimum SCR inlet temperature 330 degC on LDO.

| Circuit              | Flow kg/s | Outlet pressure bar(a) | Outlet temperature degC | Drum pressure bar(a) | Design pressure bar(a) / temperature degC |
|----------------------|-----------|------------------------|-------------------------|----------------------|-------------------------------------------|
| HP superheater       | 100.15    | 187.0                  | 600.0                   | 195.0                | 210 / 610                                 |
| Reheater             | 117.49    | 37.6                   | 600.0                   | -                    | 50 / 610                                  |
| IP superheater       | 17.34     | 40.3                   | 330.0                   | 41.8                 | 50 / 350                                  |
| LP superheater       | 11.90     | 5.40                   | 250.0                   | 5.9                  | 10 / 300                                  |
| Condensate preheater | 141.87    | 6.9                    | 153.2                   | -                    | 25 / 180                                  |

| Section        | Gas in degC | Gas out degC | Duty MW |
|----------------|-------------|--------------|---------|
| HPSH / RH      | 640.0       | 450.6        | 176.83  |
| HPEVAP         | 450.6       | 371.6        | 71.38   |
| IPSH           | 371.6       | 366.9        | 4.19    |
| HPECO2         | 366.9       | 295.1        | 63.71   |
| IPEVAP         | 295.1       | 261.0        | 29.89   |
| LPSH           | 261.0       | 258.2        | 2.43    |
| HPECO1 / IPECO | 258.2       | 198.7        | 51.64   |
| LPEVAP         | 198.7       | 166.2        | 27.91   |
| CPH            | 166.2       | 80.2         | 73.12   |

Stack temperature 80.2 degC at SRC. Condensate preheater inlet temperature is kept at not less than 50 degC (acid dew point margin, LDO) by a 2 x 100 % CPH recirculation pump (IEC); at SRC the mixed inlet from the condensate and the fuel gas heater return is 30.9 degC before recirculation.

## 3. Water/steam quality required

Per IEC-ALP-REQ-001 section 4 (IAPWS TGD and VGB-S-010): all-volatile treatment (AVT(O)) with trisodium phosphate provision for the HP drum; continuous blowdown 0.5 % maximum at start-up, 0 % in normal operation.

## 4. Weights and transport

| Item                                                         | Data                                            |
|--------------------------------------------------------------|-------------------------------------------------|
| Heaviest pressure-part module (HP evaporator)                | 185 t, 24.0 x 3.6 x 26.5 m (shipped horizontal) |
| HP drum                                                      | 96 t, 14.0 m long, OD 1.9 m                     |
| Total HRSG weight (pressure parts, casing, steel, catalysts) | 6,400 t                                         |